Shape-Adjoint Design of a Curved Conformal 2.4 GHz Patch Antenna on a Fixed 1.6 mm Board with a Differentiable Open-Radiator FEM

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Abstract

We design a probe-fed 2.4 GHz patch antenna wrapped on a cylinder of radius 100 mm, keeping the board a standard uniform 1.6 mm FR-4 shell and changing only the etched copper. The design tool is a differentiable three-dimensional Nédélec finite-element method (FEM) on unstructured tetrahedral meshes, whose discrete shape adjoint passes through the full open-radiator forward model (box UPML absorber, lossy substrate and a lumped feed port whose position and width move), at one LU factorization per frequency. Thirty-eight smooth copper degrees of freedom (outline splines, corner bumps, feed inset and width) are applied by an exact on-cylinder morph, so the board stays 1.6 mm thick. A mesh-refinement study shows that the antenna's resonance is not converged on any affordable mesh: it rises under every h- and p-refinement, and a design optimized on a coarse mesh resonates about 45–48 MHz higher with second-order elements. That detune is close to a pure frequency translation, so we correct it by frequency space mapping and verify on the finest computed model, second-order elements on the second mesh level (P2/L1). There the design reaches a worst in-band $|S_{11}|$ of -10.78 dB and a 94.9 MHz -10 dB band (2.3938–2.4887 GHz) that covers the 2.400–2.4835 GHz ISM band. On the finest first-order mesh it reaches -8.80 dB. Richardson estimates for the initial antenna place its converged resonance a further 12.7–48.4 MHz up; the design has tracked the initial antenna's shifts to within 3 MHz, and every estimate exceeds the 6.2 MHz margin at the lower band edge, so the converged design probably misses at 2.400 GHz. The feed is an idealized 4.3 mm port strip.

1 Introduction

Gradient-based inverse design has become routine in nanophotonics, where differentiable solvers such as CEVICHE (Hughes et al., 2019), the Meep adjoint solver (Hammond et al., 2022), SPINS-B (Su et al., 2019) and FDTDX (Mahlau et al., 2026) optimize a permittivity density on a structured grid. The same density approach has been carried to metallic antennas: Hassan et al. (2014b) optimize the conductivity of a microstrip patch on a finite-difference time-domain (FDTD) grid with an adjoint gradient and more than 20,000 design variables. Shape derivatives of scattering parameters for finite-element models with ports are older still (Nikolova et al., 2004a; Webb, 2002; Webb and Akel, 2000) and are available in commercial tools (Ansys, Inc., 2025; COMSOL AB, 2023).

This paper uses a differentiable finite-element solver, geode-fem, to design a specific antenna under manufacturing constraints and asks how far the result can be trusted. The antenna is a 2.4 GHz probe-fed patch bent around a cylinder of radius $R = 100$ mm. The board is a uniform 1.6 mm FR-4 shell and only the copper changes, so a design is a copper outline and a feed position

that a board house could etch on standard stock. The body-fitted tetrahedral mesh follows the curved conductor exactly, and the design variables move it.

Two findings shape the paper. First, the forward model is not converged on any mesh we can afford: the resonance of the initial antenna rises by 85 MHz from the coarsest to the finest computed model, and the order of convergence is falling along the ladder, so the converged resonance is bounded below but not above (Section 4.1). A design optimized on a coarse mesh therefore lands off band on a finer one. Second, that discrepancy is close to a pure frequency translation, which a frequency-only form of space mapping (Bandler et al., 1994, 2004) corrects with one or two refits (Section 5).

The contributions are:

- A discrete shape adjoint for a radiating antenna model on unstructured tetrahedra: Nédélec elements, a box UPML, a lossy substrate and a lumped port whose position and width are design variables, with element Jacobians from forward-mode automatic differentiation and one LU factorization per frequency shared by the forward and adjoint solves (Section 3). The gradient matches central finite differences of the full pipeline to a relative 2.2×10^{-5} or better on every run.
- A copper-only design parametrization on a fixed curved board: 38 smooth degrees of freedom applied by an exact morph that keeps every substrate node on its own cylinder (Section 3.3).
- A design verified against refinement: worst in-band -10.78 dB and a 94.9 MHz -10 dB band on the finest computed model, with the finer first-order mesh, an independent sibling mesh and an extrapolation of the converged response reported alongside, including the likely miss at the lower band edge (Section 6).
- Two ablations: without the frequency correction the same procedure gives -4.76 dB on the finest model, and an idealized design that also varies the board thickness reaches -9.16 dB there (Section 7).

2 Related Work

Finite-element S-parameter sensitivities with ports. Design sensitivities of the scattering matrix for tetrahedral edge elements were derived by Webb and Akel (2000), extended to the frequency response by Webb (2002) and used in gradient optimization of microwave devices by Nair and Webb (2003). Adjoint and self-adjoint formulations for S-parameter sensitivities are reviewed by Nikolova et al. (2004a) and Nikolova et al. (2006b), and continuum shape sensitivity for high-frequency devices is given by Choi et al. (2011). HFSS computes analytic S-parameter derivatives with respect to geometric parameters by reusing the LU factors (Ansys, Inc., 2025; Vardapetyan et al., 2008), and COMSOL documents an adjoint sensitivity method and shape optimization of a waveguide filter and a diplexer on a deformed mesh (COMSOL AB, 2023; Jensen, 2022). These examples are closed waveguide devices with a few geometric parameters. For antennas, Ghassemi et al. (2013) evolve a microstrip antenna with an FEM adjoint sensitivity and a small set of control vertices. Our model differs in being an open radiator (box UPML, lumped port, lossy FR-4) with 38 smooth boundary degrees of freedom and a moving feed.

Automated shape differentiation. Shape derivatives of finite-element models can be generated automatically: in UFL and Firedrake (Ham et al., 2019, 2020; Paganini and Wechsung, 2021) and in NGSolve (Gangl et al., 2021). Wang and Anderson (2011) give a discrete Maxwell shape adjoint on unstructured meshes in two dimensions, in the time domain and without ports. geode-fem obtains

the element Jacobians by forward-mode differentiation of its closed-form element kernels inside a tensor stack (Tracel AI, 2024), rather than by symbolic shape calculus.

Structured-grid design of metallic antennas. Density-based topology optimization of conductors on a fixed grid goes back at least to Erentok and Sigmund (2011). Hassan et al. (2014b) and Hassan et al. (2014a) optimize the patch and ground plane of coax-fed microstrip antennas with an FDTD adjoint, and Nikolova et al. (2004b, 2006a) give FDTD sensitivities with respect to shape parameters. A plain Yee grid staircases a curved conductor, which produces $O(1)$ field errors next to the boundary (Cangellaris and Wright, 1991; Häggblad and Runborg, 2014), but structured grids need not staircase: locally conformal FDTD has handled curved PEC for decades (Dey and Mittra, 1997; Mittra and Yu, 2000), including uniformly stable second-order schemes (Benkler et al., 2006; Zagorodnov et al., 2003). In differentiable photonics, subpixel-smoothed projection gives the Meep density adjoint subpixel-accurate binarized boundaries (Hammond et al., 2025; Romano et al., 2026), though the material-grid adjoint excludes metallic media (Meep developers, 2026); Tidy3D offers autograd shape gradients and conformal PEC averaging in its forward solver (Flexcompute, Inc., 2026); and Hooten et al. (2025) and Liu and Bonar (2024) map shape parameters to a fixed grid. Precomputed numerical Green's functions make pixelated antenna design fast without gradients (Sun et al., 2025). We did not find a conformal FDTD adjoint for the shape of a curved conductor, and we do not compare against any of these methods here (Section 8).

Integral-equation shape sensitivity. Boundary- and integral-equation methods carry recent PEC shape-sensitivity work: time-domain boundary-integral shape derivatives (Takahashi, 2025), freeform tubular scatterers (Arens et al., 2026) and isogeometric coaxial baluns (Balouchev et al., 2024).

Space mapping. Space mapping aligns a cheap coarse model with an accurate fine model and optimizes the coarse one (Bandler et al., 1994, 2004). We use its simplest form, a frequency shift fitted between two discretizations of the same solver.

Position of this work. The closest prior work is the FEM S-parameter sensitivity line above, which already reuses factorizations and differentiates with respect to geometry. What we add is the combination, for an open radiator, of a box UPML, a lumped port whose position and width move, a lossy substrate and a smooth many-parameter copper basis on a curved board, together with a design verified against mesh and order refinement. We do not claim a first EM shape adjoint, and we make no performance comparison with structured-grid methods.

3 Method

3.1 The open-radiator forward model

The forward problem is the driven frequency-domain curl-curl equation, discretized with Nédélec edge elements of order $p = 1$ or $p = 2$ on an unstructured tetrahedral mesh. Lengths are in millimetres and the solver works with $\omega = k_0 = 2\pi f/c$ in rad/mm. The assembled system is the complex-symmetric pencil

$$A(\omega)\,\mathbf{x} = \Big[\, K(\nu) - \omega^2 M(\varepsilon) + \tfrac{j\omega}{Z_s}\, S_p \,\Big]\,\mathbf{x} = \mathbf{b}, \tag{1}$$

with curl-curl stiffness K, mass matrix M and lumped-port surface term S_p. The radiating boundary is a box UPML: per-element tensor weights Λ enter both K (as $\nu = \Lambda^{-1}$) and M (as $\varepsilon = \varepsilon_r \Lambda$),

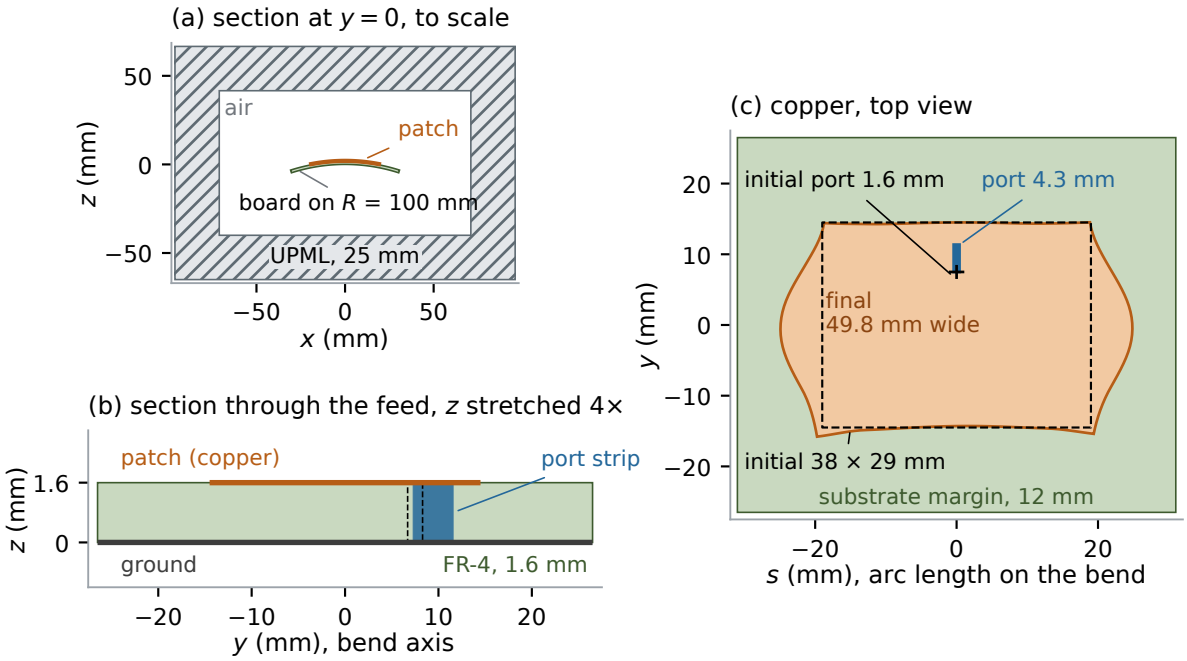


Figure 1: The fixture and the final copper design. (a) Section at $y = 0$, to scale: the FR-4 board (green) on the $R = 100\,\text{mm}$ cylinder with the patch on top, inside the air box and the 25 mm box UPML (hatched). The bend is blended to zero at the air-box wall, so the UPML stays a rectangular box. (b) Section through the feed at $s = 0$, with the vertical axis stretched four times: patch, 1.6 mm FR-4, ground, and the lumped-port strip across the gap, final (filled) and initial (dashed). (c) Top view in the chart (s, y): the initial $38 \times 29\,\text{mm}$ patch (dashed), the final copper outline (filled), and the initial and final port strips. The board outline and thickness are the same in both. Drawn from `baseline_2g4_L1.toml` and `geometry/rbD_L1_copper_38_sm45.json`.

with stretching $s = 1 - j\sigma_0(d/w)^2/\omega$ per axis. The FR-4 has $\varepsilon_r = 4.4$ and $\tan\delta = 0.02$, and the conductors are perfect electric conductors (PEC). The port is a rectangle of width w along y and length $l = 1.6\,\text{mm}$ across the substrate gap, with sheet impedance $Z_s = R\,w/l$ and $R = 50\,\Omega$ (entered in the solver as $50/\eta_0$). The reflection coefficient follows from the port voltage V:

$$S_{11} = \frac{Z - R}{Z + R}, \qquad Z = \frac{V}{I}, \qquad I = \frac{2V_{\text{inc}} - V}{R}. \tag{2}$$

Every evaluation checks passivity, $|S_{11}| \le 1$.

3.2 Shape adjoint with one factorization per frequency

The design moves mesh node coordinates X, so the gradient needs $\partial K/\partial X$, $\partial M/\partial X$, $\partial S_p/\partial X$ and $\partial \mathbf{b}/\partial X$. geode-fem evaluates the closed-form element kernels in first-order dual arithmetic: seeding the tangent of one node coordinate returns the exact partial derivatives of every local matrix entry with respect to it. The UPML weights Λ are held fixed under node motion, so the tangent flows through geometry only. Because $A(\omega)$ is complex symmetric, the adjoint system uses the same operator, and one LU factorization per frequency serves both the forward and the adjoint solve. The nodal gradient at each frequency is $\lambda^{\top}(\partial A/\partial X)\,\mathbf{x}$ plus the explicit dependence of the port drive and of the port-voltage functional on the geometry.

When a design variable moves the feed, the port faces move: w, l, the drive and the voltage functional all depend on X. The moving-port terms are differentiated together with the volume

Table 1: The 38 copper degrees of freedom. Units are per unit of the coefficient. The board thickness has no degree of freedom.

Group	Count	What moves
Outline	32	8 clamped cubic B-splines per patch edge ($\pm y$, $\pm s$); each moves its edge outward in the surface by 1 mm, ramped in from the feed band and decaying over the 12 mm substrate margin
Corners	4	one bump per corner, moving it outward along the bisector by 1 mm (negative values chamfer), support $1 - \mathrm{smoothstep}(d/6\,\mathrm{mm})$
Feed	2	the port's position along y (its inset from the $+y$ edge) and its width w along y, each 1 mm

terms, and the port is rebuilt on every moved mesh. A unit test of the moving-feed gradient on a curved mesh, with a current source that varies across the port so that the explicit term is not negligible, matches central finite differences to below 10^{-8}; without the explicit term it misses by a relative error of about 3.

3.3 Copper design freedom on a fixed curved board

The board is not a design variable. Each substrate node is described in a chart (s, y), with $s = R\,\mathrm{atan2}(x, z - z_{\mathrm{axis}})$ the mid-plane arc length and y the bend axis. A design applies a chart displacement $(\Delta s, \Delta y)$ and moves the node to angle $\phi_0 + \Delta s/R$ and height $y_0 + \Delta y$ at its own base radius. The radial column through the slab therefore rotates together, the FR-4 stays a uniform 1.6 mm shell, the patch stays on the $R + h/2$ surface and the ground on $R - h/2$. Every run records the largest change of any substrate node's radius ($\leq 1.4 \times 10^{-14}$ mm) and the slab thickness (1.600 mm everywhere). Air nodes follow a harmonic (P1-Laplace) extension of the motion; the UPML shell and the outer boundary are fixed. The design gradient chains the nodal adjoint gradient through the exact Jacobian of this map.

Table 1 lists the 38 degrees of freedom. They are smooth functions of the chart alone, so the same coefficient vector defines the same copper on every mesh of the family, which is what lets a design be carried between discretizations. The basis cannot open a hole in the patch: slots (U-slot, E-patch) change the topology of the conductor and are outside this parametrization.

3.4 Objective, optimizer and guards

The objective is a soft maximum of $|S_{11}|^2$ over $n_f = 5$ band frequencies,

$$G(\mathbf{c}) = \frac{1}{\beta} \ln \sum_f \exp\big(\beta\, |S_{11}(f; \mathbf{c})|^2\big), \qquad \beta = 25, \tag{3}$$

over the coefficient vector $\mathbf{c}$. It is minimized by L-BFGS (memory 8) with Armijo backtracking and a cap of 1 coefficient unit per step. A per-element guard rejects any trial step that would shrink a tetrahedron's signed volume below 0.25 of its base value. In the runs reported here the guard, not the gradient, ends the descent: every run stops at the volume budget (or on a plateau) with the gradient norm still nonzero, so the designs are guard-limited rather than stationary.

Before and after each run the full pipeline gradient is compared with a central finite difference along the steepest-descent direction and along random directions in all 38 coefficients. The worst relative error over all runs is 2.2×10^{-5} (tolerance 5×10^{-3}); a per-group check of the outline, corner, inset and width coefficients on the two coarsest meshes passes to 1.9×10^{-7}. The adjoint is implemented for $p = 1$ only, so optimization runs at $p = 1$ and second-order elements enter through verification.

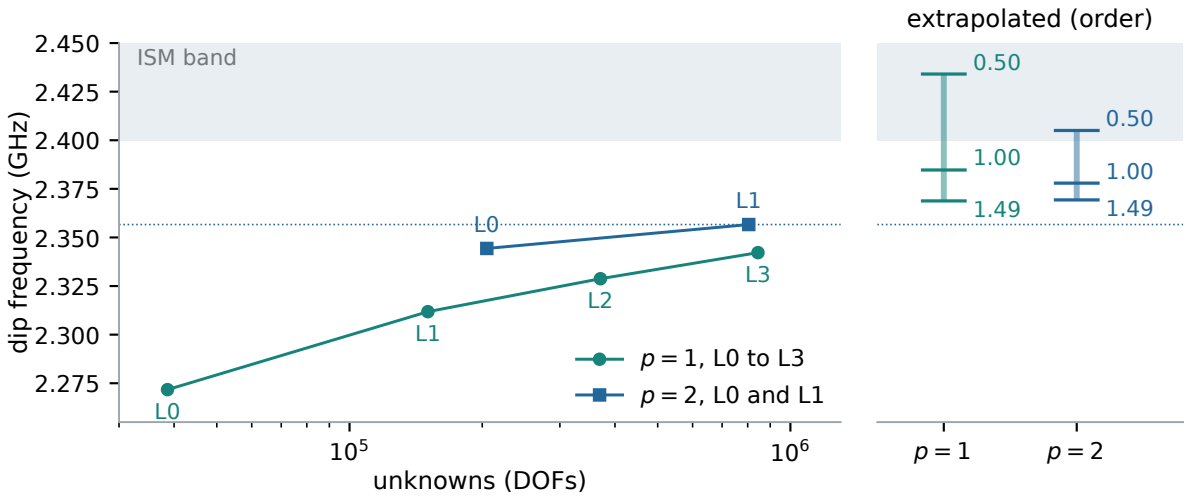


Figure 2: Resonance (the $|S_{11}|$ dip) of the initial bent patch under refinement, against the number of unknowns. Left: $p = 1$ on L0–L3 and $p = 2$ on L0 and L1. The dip rises under every refinement step; the dotted line is the finest computed value, 2.3566 GHz at P2/L1, a lower bound on the converged dip under that monotone trend. Right: Richardson extrapolations of the $p = 1$ ladder and of the $p = 2$ pair at the three orders shown (1, the local order of the finest triple, and a least-squares slope). They are estimates under assumed orders, not bounds. The shaded band is the ISM band. Data: `convergence.toml`.

4 Fixture and Discretization

Antenna. The initial antenna is a 38×29 mm PEC patch (width 38 mm along s, resonant length 29 mm along y) on 1.6 mm FR-4 with a 12 mm substrate and ground margin, fed by a lumped-port strip 1.6 mm wide inset 7.0 mm from the $+y$ edge (Figure 1). The flat mesh is wrapped onto a cylinder of radius 100 mm about an axis parallel to y; the 62 mm substrate then subtends ± 0.31 rad, with a sagitta of 1.8 mm at the patch edge. A 40 mm air margin separates the board from a 25 mm box UPML. The target is the 2.400–2.4835 GHz ISM band at -10 dB.

Discretization ladder. Four graded meshes L0–L3 carry the same geometry with 38,650, 150,561, 370,780 and 844,238 edges, and second-order elements run on L0 and L1 (204,824 and 804,460 unknowns). We write P2/L1 for $p = 2$ on L1. The UPML conductivity $\sigma_0 = 1$ was chosen from a ladder of σ_0 and shell thickness on L1, where the residual sensitivity of S_{11} to the absorber setting is $|\Delta S_{11}| \leq 0.021$.

Bending. At this radius the bend barely moves the resonance: the bent dip lies 1.0 to 1.6 MHz below the flat dip on every level. The curvature matters for representing the conductor and for the design freedom, not as a large detune.

4.1 The forward model is not converged

Figure 2 shows where the initial antenna resonates on each model. The dip rises under every h- and p-refinement step, from 2.2717 GHz on L0 to 2.3422 GHz on L3 and 2.3566 GHz at P2/L1. At a similar number of unknowns P2/L1 lies 14.5 MHz above L3, so the second-order model is the more accurate one, and the finest computed dip bounds the converged dip from below. The upper end is not determined. The local order of the dip frequency falls along the $p = 1$ ladder, from 1.18 on L0–L1–L2 to 0.50 on L1–L2–L3, so the levels are not in the asymptotic range. Extrapolating the

$p = 2$ pair gives 2.3693, 2.3779 and 2.4050 GHz at orders 1.49, 1 and 0.50, that is 12.7, 21.3 and 48.4 MHz above P2/L1. Order 1 is the rate expected from the $r^{-1/2}$ field singularity at the patch edge, but we have no measurement that selects it.

For design this has two consequences. A design centred on any computed model will resonate higher on a finer one. And since the in-band worst case of the initial antenna improves with fidelity but is still −2.42 dB at P2/L1, the optimization problem is real at every fidelity we can compute.

Costs set the ladder. On a 28-core workstation an L1 forward solve takes about 44 s and a band gradient at three frequencies 67 s, at 6–8 GiB peak memory. A P2/L1 solve takes 753 s and 132 GiB on a 32-vCPU cloud host, and an L3 solve about 8 min and 68 GiB. We therefore optimize on L1 or L2 and verify on L3 and P2/L1.

5 Design Procedure

Coarse optimization. A cold start on L1 optimizes the band directly and reaches a worst-of-band −10.04 dB on L1 after 24 steps, ending when the volume guard binds in a tetrahedron between the feed and the $+y$ edge. With P2/L1 the same copper gives only −4.76 dB (Figure 3b): the design resonates higher on the finer model.

Frequency space mapping. We model the fine response as the coarse response shifted in frequency, $S_{\text{fine}}(f) \approx S_{\text{coarse}}(f - \delta)$, and fit δ by least squares over the fine model's sampled points, with the coarse response interpolated on a dense coarse sweep. For the cold L1 optimum $\delta = 47.85$ MHz, and the root-mean-square complex misfit falls from 0.23–0.38 unshifted to 0.018–0.025 shifted, over the copper designs fitted. The shift is within 3 MHz of the initial patch's own L1-to-P2/L1 shift (44.8 MHz), so the copper motion adds almost no detune of its own. The corrective step re-optimizes on the coarse model for the band shifted down by δ and checks the result at P2/L1. One refit was needed: the first corrected design gave −9.96 dB at P2/L1, the refitted shift was 44.94 MHz, and the second corrected design is the final one.

Feed re-basing. Moving the feed toward the $+y$ edge compresses the 5–7 mm of mesh between them, and the cold runs on every level stop there at the volume guard. We then regenerate the mesh with the port at its moved position and width and carry the design over, shifting the two feed coefficients by the change of the fixture's port. From an original mesh this carries the copper exactly. Between two regenerated meshes it does not: the feed coefficients' support also reaches the $+y$ edge there, and the re-base drops that motion, so the target mesh carries a slightly different copper. We report that difference wherever it applies (Section 6). After re-basing, the optimizer widens the patch from 38 to about 49.8 mm and the port from 1.6 to about 4.3 mm, until the guard binds in the substrate margin 5–6 mm outside a radiating edge.

Table 2 lists the chain that produced the final design.

6 Results

Finest computed model. At P2/L1 the final design has a worst $|S_{11}|$ of −10.78 dB over nine points spanning the ISM band, and the 41-point sweep gives a contiguous −10 dB band from 2.3938 to 2.4887 GHz, 94.9 MHz wide, which covers the whole 83.5 MHz ISM band (Figure 3a, Table 3). The band is centred to 0.5 MHz. The margins are small: the −10 dB band starts 6.2 MHz below 2.400 GHz and ends 5.2 MHz above 2.4835 GHz. This verification runs on the mesh the design was optimized on, so the verified copper is the optimized copper (they agree to about 10^{-3} mm), and its minimum volume ratio is the guard value 0.250. As built, the patch is 28.80 mm long and 49.75 mm

Table 2: The optimization chain of the final design. "Worst" is the worst of the band points on the band each run optimized (shifted down by δ where stated), on the mesh it ran on. All runs are at $p = 1$ on L1 meshes; "re-based" means the mesh was regenerated at the moved port. The wall times are on cloud hosts of 16 or 32 vCPU.

Step	Start	δ (MHz)	Worst (dB)	Steps	Terminal	Wall
1 cold	initial patch	0	$-1.47 \to -10.04$	24	guard (feed)	0.8 h
2 re-based	step 1	0	$-9.97 \to -10.03$	4	plateau	0.5 h
3 corrected	step 2	47.85	$-4.75 \to -10.25$	26	guard (widening)	0.8 h
4 refit, re-based	step 3	44.94	$-10.26 \to -10.78$	9	guard (widening)	0.7 h

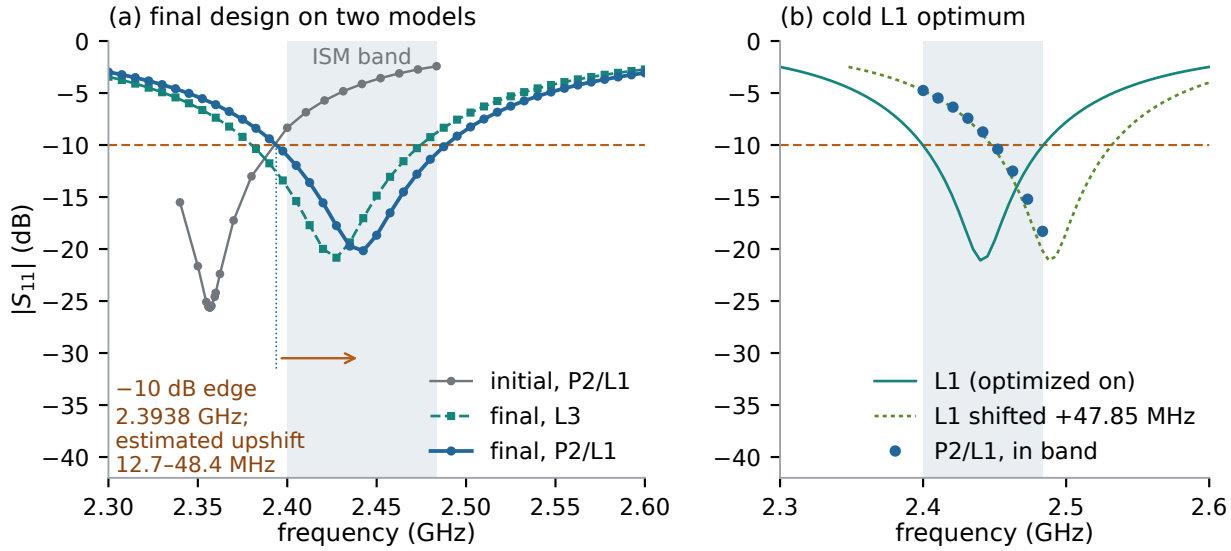


Figure 3: (a) Return loss of the final design at P2/L1 (blue, 41 points) and on L3 (teal, dashed, 41 points), and of the initial patch at P2/L1 (gray, sampled near its dip and at nine in-band points). The dashed orange line is −10 dB and the shaded band is the ISM band. The arrow starts at the lower −10 dB edge at P2/L1 and has the length of the largest extrapolated further upshift (48.4 MHz); the smallest is 12.7 MHz (Section 4.1). (b) The cold L1 optimum (step 1 of Table 2) on L1, where it was optimized, its nine in-band points at P2/L1, and the L1 curve shifted by the fitted 47.85 MHz. Data: `copper/verify/` and `convergence/`.

wide at the feed, the port strip is 4.27 mm wide and 5.07 mm from the $+y$ edge, and the board is 1.600 mm thick everywhere.

Finest first-order mesh. On L3 the same copper gives −8.80 dB and covers 89% of the band. Its −10 dB band is about as wide (94.6 MHz) but sits 14.4 MHz lower, close to the initial patch's 14.5 MHz L3-to-P2/L1 offset. With 95 MHz of bandwidth against an 83.5 MHz band there are only about 11 MHz of slack, so a design centred for one model is off centre on the other. The L3 mesh of the widened patch has thin air elements next to the slab margin, which the default air morph compresses to a volume ratio of 0.10; moving the air on cylinders instead raises it to 0.24 and changes $|S_{11}|$ by 0.01 dB, so the L3 shortfall is not a mesh-quality artifact.

Where the converged design probably falls. P2/L1 is the better reference: the dip rises under refinement and P2/L1 lies above L3 at a similar size, so L3 misses on the less accurate side.

Table 3: Fresh forward verifications. "Worst" is over the nine in-band points; the band columns come from dense sweeps (41 points, 2.30–2.60 GHz; 21 points, 2.35–2.55 GHz for the related design). Coverage is the fraction of the ISM band below −10 dB. "Vol" is the smallest tetrahedron volume ratio on the verifying mesh.

Design	Model	Worst (dB)	−10 dB band (GHz)	Coverage	Vol
Initial patch	P2/L1	−2.42	none in band	0%	1.00
Final	P2/L1	−10.78	2.3938–2.4887	100%	0.250
Final	L3	−8.80	2.3801–2.4746	89%	0.241
Final, sibling mesh (−0.09 mm)	P2/L1	−10.35			0.234
Related L2-chain design (−0.47 mm)	P2/L1	−10.10	2.3993–2.4936	100%	0.299
Related L2-chain design (−0.47 mm)	L3	−9.25			0.228

The same data put the converged response above P2/L1. The extrapolated further rise of the initial patch's dip is 12.7–48.4 MHz (Section 4.1), and the copper design's coarse-to-P2/L1 shift tracks the initial patch's to within 3 MHz, so a similar further rise is expected. Every extrapolated rise exceeds the 6.2 MHz margin at the lower band edge. The design meets the −10 dB target across the ISM band on the finest computed model, and on the converged estimate it probably misses at 2.400 GHz.

Sibling mesh. An independently regenerated L1 mesh at the same port gives −10.35 dB at P2/L1, about 3 MHz higher. This is not a pure remeshing test: the re-base onto that mesh also shortened the patch by 0.09 mm at the feed (Section 5), and a shorter patch resonates higher, so the 3 MHz mixes remeshing with a small design change.

A related design from an L2 chain. A second chain optimized on L2 (shift 29.33 MHz) reaches a design of the same family: patch about 28.9 × 49.0 mm, port 4.5 mm. Its verifications ran on regenerated meshes whose re-base shortened the $+y$ edge by 0.47 mm at the feed, so they verify a related design, not the L2 optimum: −10.10 dB and a 94.3 MHz band covering the ISM band at P2/L1, with 0.7 MHz of margin at 2.400 GHz, and −9.25 dB on L3. No copper design has been verified on L3 and at P2/L1 exactly as optimized on L2.

Bandwidth in context. The closed-form estimate of Jackson and Alexopoulos (1991) for a flat rectangular patch on this substrate, with the FR-4 loss tangent included and a voltage standing-wave ratio of 2, gives about 61 MHz for the initial 38 mm patch and 69 MHz for the 49.8 mm final patch. The FEM bandwidths are larger: 84 MHz for the 38 mm cold optimum on its own L1 mesh, and 94.9 MHz for the final design at P2/L1. The closed form ignores the feed and the curvature, and we have not resolved the difference. Within the FEM results, the increase from 84 to 95 MHz comes with the two standard bandwidth measures for a single patch, a wider patch (lower radiation Q) and a wider feed (lower feed inductance), rather than any added structure.

Cost. The final chain took 2.8 h of optimization on cloud hosts (Table 2). The last run made 125 driven solves in 40 min; its first gradient (five frequencies, forward and adjoint on one LU each) took 104 s at 6.2 GiB. The 41-point P2/L1 sweep of the final design took 5.4 to 7.7 min per frequency at 151 GiB peak memory on a 32-vCPU host, in three shards.

7 Ablations

Without the frequency correction. The cold L1 optimum (step 1) is the same procedure without space mapping. It reaches −10.04 dB on L1 and −4.76 dB at P2/L1. With the fitted shift

and one refit, the final design reaches −10.78 dB at P2/L1. On this comparison the correction, not additional design freedom, accounts for most of the 6 dB. A second coarse model gives the same picture with a smaller shift: a design optimized on L2 with a 30 MHz hedge gives −9.92 dB at P2/L1, and the fitted L2 shift is 29.33 MHz.

Varying the board thickness. In an idealized ablation the design freedom also moves the patch radially, so the FR-4 under the patch thickens from 1.6 mm to about 3.7 mm on average (up to 4.7 mm), with the feed left at the bottom of a 1.6 mm crater. Such a board cannot be made from standard stock. Optimized on L2 without a frequency correction, the best such design reaches −14.07 dB on L2, −12.47 dB on L3 and −9.16 dB at P2/L1, missing the target by 0.84 dB at 2.400 GHz after a 4.9 dB loss from L2 to P2/L1. The same freedom optimized on L1 loses 8.4 dB to P2/L1 (−14.12 to −5.69 dB). These designs also detune by about 75 MHz between models, more than the copper designs, because the thickness change adds a detune of its own. The copper-only design meets the target at P2/L1 with less freedom; on this evidence that is due to the frequency correction removing the transfer loss, not to copper being a better design variable than thickness.

What the data do not separate. We have not compared the adjoint-driven optimizer with a gradient-free search or with a low-dimensional parametric sweep of patch size and feed position on the same models. The final design widens the patch and the feed, which a parametric sweep could also explore, so the result does not show that the 38-parameter basis or the shape gradient is necessary for this antenna.

8 Limitations and Future Work

Feed. The feed is a uniform lumped-port strip across the 1.6 mm gap. The final design's 4.3 mm port is a strip of that width, not a coaxial pin (an SMA centre pin is 1.27 mm). A real feed would need a wide post or strip, or a matching structure presenting the same port, and none is modelled or verified. The copper outline is manufacturable on a standard board; the feed is not shown to be.

Convergence and centring. P2/L1 is the finest model and it is not converged. Centring the design on the extrapolated converged response would take another space-mapping step with a larger shift, but the 12.7–48.4 MHz spread of the extrapolations exceeds the design's 11 MHz of total slack, so the extrapolation would have to be narrowed first, for example with a second-order L2 solve. The adjoint is first-order only; a $p = 2$ shape adjoint would let the optimizer run on the more accurate model directly.

Scope. The objective is impedance match only; radiation pattern, gain and efficiency are not computed. Conductors are PEC, so copper loss is not modelled. The design is one antenna on one radius and has not been fabricated or measured.

Comparison with structured-grid methods. A natural question is how a structured-grid design method would fare on the same curved board at matched accuracy. A Meep (Oskooi et al., 2010) model of this fixture has been set up and piloted, but its runs are not yet converged, so we make no cost or accuracy comparison. Our working hypothesis is that a body-fitted mesh reaches a given accuracy on the curved conductor at lower cost than a staircased grid; testing it needs a converged FDTD reference, ideally including a conformal FDTD scheme (Dey and Mittra, 1997; Zagorodnov et al., 2003), and is left to future work.

9 How geode-fem Was Designed and Built

geode-fem implements the whiteroom L1–L4 operator specification of Crutcher Dunnavant (Dunnavant, 2026). He also gave design input on the solver's operator layer and is the author of 36 commits, made in June 2026, that migrated the code to Rust 2024, added the Metal (Apple GPU) backend, restructured the core crate and the examples into their present layout, and created the geode-util crate. The code, the experiments and the drafts of this paper were otherwise produced by frontier AI models working as role-separated agents: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). Codex reviewed the figures and page layout of a draft of this paper. The agents were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by a shared Squad room.

The five pull requests that produced this paper's results (the fixture, the smooth basis, the refinement study, the thickness ablation and the copper design) each received an independent review by a separate agent session, which recomputed reported numbers from the committed artifacts with its own scripts, reproduced forward points and mutation-tested the regression pins. Several claims were corrected because of those reviews. Three examples follow. The converged resonance of the initial patch was first bracketed 29–43 MHz below the band from a single fitted slope; solving each level triple for its own order showed the order falling, and the result is now a lower bound with an undetermined upper end (Section 4.1). A review found that re-basing the feed between two regenerated meshes changes the copper by 0.09 or 0.47 mm, so the sibling-mesh check and the L2-chain results are reported as perturbed or related designs (Section 6). And a review required the paper to state which way the converged answer probably falls, rather than calling it unknown.

Counts from repository history, up to this revision: the paper and its benchmark artifacts changed through 13 merged pull requests from 2026-07-20 and 30 commits. Of these commits, 26 carry a Claude co-author trailer (Opus 19, Fable 7); the other 4 carry no trailer and contain about 9% of the added text lines. The human author directed the research, made every scoping and claim decision, and takes responsibility for the content.

10 Reproducibility

Every result in this paper is read from a committed, scripted artifact.[1] Table 4 lists them, relative to `benchmarks/patch_antenna_conformal/`. Each TOML records the exact command, the mesh SHA-256 and the commit that produced it, and regression tests pin the reported values. The meshes are regenerated from YAML recipes in `reference/gmsh/` with gmsh. The figure scripts in `figures/src/` read the artifacts directly, and `provenance.md` in this paper's directory maps every number to its source field.

11 Conclusion

A differentiable finite-element model of a curved, PML-terminated patch antenna, with a feed that moves, gives exact shape gradients at one factorization per frequency, and with them a copper-only design on a standard 1.6 mm board reaches −10.78 dB across the ISM band on the finest model we can compute. The harder part was the discretization. The resonance is not converged on any affordable mesh, a coarse-mesh optimum lands 45–48 MHz off on a finer one, and the margin left at the lower band edge (6.2 MHz) is smaller than every estimate of the remaining shift. A frequency-only space mapping removed most of the transfer loss, and the same data say the converged design

[1] The geode-fem source is public at https://github.com/rjwalters/geode-fem. Archival DOI to be added at deposit.

Table 4: Artifacts behind each result, relative to `benchmarks/patch_antenna_conformal/`.

Result	Artifact
Fixture and flat-versus-bent baseline	`baseline_2g4_L0.toml`, `baseline_2g4_L1.toml`
Refinement study, absorber ladder, cost (Section 4.1, Figure 2)	`convergence.toml`, `convergence/`, `upml_ladder_L1.toml`
Copper optimizations (Table 2)	`copper/*.toml`, `copper/README.md`
Verifications and bands (Table 3, Figure 3)	`copper/verify/`, `copper/g2_summary.toml` from `scripts/copper_g2.py`
Frequency shift fit	`scripts/copper_shift_fit.py`
Geometry of the designs (Figure 1)	`geometry/rbD_L1_copper_38_sm45.json` and the other `geometry/*copper*.json`
Thickness ablation (Section 7)	`nested_dof/`, `nested_dof/verify/`, `scripts/nested_dof_g2.py`
Generator and tests	`crates/geode-core/examples/patch_conformal_nested_dof.rs`; `crates/geode-core/tests/patch_conformal_copper.rs`

probably misses at 2.400 GHz. For a gradient-designed antenna, the refinement study is part of the result.

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